

Evaluation of Radiation-Hardened SRAM Bitcell Architectures for Space-Based AI Inference

John Gannon, Arnur Maratov, and Yoran Wong

Abstract—The deployment of machine-learning inference workloads in space environments demands memory subsystems resilient to single-event upsets (SEUs) caused by high-energy particle strikes. This work presents a simulation-based evaluation of four SRAM bitcell topologies — the standard 6T cell, the 12T DICE cell, the 14T RSP cell, and the 10T Quatro cell — implemented in the GPDK 45 nm process ($V_{DD} = 1.1$ V, $T = 27^\circ\text{C}$) using Cadence Virtuoso ADE L and the Spectre simulator. Critical charge (Q_{crit}) under both single-node and dual-node upset conditions, static noise margin, read access time, and leakage power are characterized and compared at minimum sizing. The DICE cell is identified as the most suitable architecture for space AI inference: it provides complete SNU immunity and layout-independent DNU Q_{crit} of 7.32 fC to 10.02 fC, while the RSP cell’s worst-case DNU Q_{crit} of 3.77 fC falls below the unhardened 6T baseline. A comprehensive parametric sweep of the DICE cell across transistor sizing, temperature, supply voltage, and process corners reveals a monotonic improvement in Q_{crit} with device width and a strong degradation with temperature, motivating active thermal management as a system-level reliability strategy for space deployment.

Index Terms—SRAM, radiation hardening by design, single-event upset, dual-node upset, critical charge, static noise margin, DICE, space computing, AI inference, 45 nm CMOS.

I. INTRODUCTION

RECENT industry announcements from NVIDIA and SpaceX signal that orbital data centers for AI inference may be the next frontier in computing [1], [2]. Although the availability of solar energy alleviates power utilization constraints, the harsh radiation environment of low Earth orbit (LEO) poses significant reliability challenges for conventional CMOS circuits. In this environment, high-energy protons and heavy ions interact with logic and memory circuits, depositing charge on potentially sensitive nodes. If the deposited charge is of sufficient magnitude, it may flip the stored state of a bistable cell, causing a single-event upset (SEU) [3]. For an AI inference engine, a flipped weight bit can silently corrupt model outputs with no visible fault indicator. As a result, the tolerance of the underlying memory fabric to both single-node upsets (SNUs) and the more demanding dual-node upsets (DNUs) becomes a critical design constraint.

Several radiation-hardening-by-design (RHBD) topologies have been proposed to address this challenge. The dual interlocked storage cell (DICE) [4] achieves theoretical SNU

immunity through a feedback-interlocked 12-transistor structure. The Quatro-10T cell [5] offers a compact alternative with selective hardening, while the RSP-14T cell [6] pursues a higher, finite SNU hardening ceiling at the cost of increased area and layout dependence.

This paper presents a systematic simulation-based comparison of the 6T baseline, 12T DICE, 14T RSP, and 10T Quatro bitcells using the GPDK 45 nm PDK. We characterize Q_{crit} under both SNU and DNU injection, and further evaluate SNM, read/write performance, and leakage power. The goal is to identify the architecture best suited for high-reliability AI inference memory in space, and to quantify the transistor count versus hardening trade-off for each topology. The most promising architecture is then subjected to a comprehensive parametric study examining how Q_{crit} , SNM, and leakage power scale with transistor sizing, operating temperature, supply voltage, and process corners.

The remainder of the paper is organized as follows. Section II reviews SEU physics and the four target topologies. Section III describes the simulation methodology. Section IV presents and analyzes the results. Section V offers a comparative discussion. Section VI concludes.

II. BACKGROUND

A. Single-Event Upset Mechanism

When a high-energy particle passes through a semiconductor substrate, it deposits energy along its track through ionization. For heavy ions, this process is direct: the particle strips electrons from silicon atoms as it traverses the material, generating electron-hole pairs at a rate characterized by the particle’s linear energy transfer (LET) [3]. Protons and neutrons, by contrast, deposit charge indirectly through nuclear reactions with silicon nuclei, producing energetic recoil ions that then ionize the surrounding lattice.

The ionization trail creates a column of free carriers in the vicinity of a reverse-biased p-n junction, such as those present at the drain of an off-state NMOS transistor storing a logic high. A combination of drift, driven by the junction electric field, and diffusion rapidly funnels these carriers onto the sensitive node, producing a transient current pulse lasting on the order of hundreds of picoseconds. Because the gate capacitance of a minimum-sized transistor in a deep-submicron process is extremely small — on the order of femtofarads — even a modest charge collection event can produce a voltage excursion large enough to overcome the restoring force of the cross-coupled inverter pair that forms the core of an SRAM cell.

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The critical charge, Q_{crit} , is defined as the minimum collected charge required to cause a permanent state change in the cell [3]. If the total collected charge $Q_{coll} < Q_{crit}$, the feedback network restores the original state before the voltage excursion propagates; if $Q_{coll} \geq Q_{crit}$, the perturbed node is driven past the metastable switching point and the cell latches into the opposite state, producing a single-event upset. Scaling to smaller technology nodes reduces both supply voltage and node capacitance, which together decrease Q_{crit} and make unprotected cells increasingly susceptible to SEU at lower particle energies.

A single-node upset (SNU) refers to the case in which charge is collected on one sensitive node, which is the dominant failure mode for well-separated circuit nodes. A dual-node upset (DNU) occurs when a single particle strike, or two nearly simultaneous strikes, disturbs two nodes concurrently. DNUs become increasingly probable as device pitch shrinks and as particle tracks grow long relative to node spacing, making DNU tolerance a meaningful design requirement for space-grade memory at advanced process nodes.

B. SRAM Topologies Under Study

As shown in Fig. 1, the four bitcells evaluated in this work span a range of transistor counts and hardening strategies. The following analysis describes the store 1 case for each bit cell architecture where individual nodes are referenced.

1) *6T Baseline*: The standard six-transistor cell, Fig. 1(a), serves as the unhardened reference point for all comparisons. It is fully susceptible to SNU, as a single particle strike on either storage node is sufficient to flip the cell. However, its compact

footprint and well-characterized behavior make it the industry baseline against which hardened topologies are evaluated.

2) *12T DICE Cell*: The DICE cell, Fig. 1(b), achieves theoretical SNU immunity through the structure of its feedback network [4]. Each of the four internal storage nodes is reinforced by two independent feedback paths, so a single-node perturbation is actively corrected before it can propagate, which holds regardless of technology node or particle LET. The critical DNU case occurs when both nodes of the same storage polarity are struck simultaneously: the Q1+Q2 pair (both storing logic 1) and the Q1B+Q2B pair (both storing logic 0) share a feedback dependency such that no independent restoring path remains active under simultaneous injection. Complementary pairs such as Q1+Q1B retain independent restoring paths and are not vulnerable to DNU by this mechanism. The resulting DNU Q_{crit} is set by the per-node drive strength of the cell rather than by physical layout separation.

3) *14T RSP Cell*: The RSP cell, Fig. 1(c), increases hardening through feedback redundancy and deliberate transistor sizing [6]. A 3:1 driver-to-pull-up width ratio strengthens the restoring current available after a strike, yielding SNU Q_{crit} values substantially above the 6T baseline. The key limitation is that DNU hardening is non-uniform: certain topologically adjacent node pairs share a common pull-down path, creating a worst-case DNU vulnerability whose severity depends critically on physical node separation in the final layout.

4) *10T Quatro Cell*: The Quatro cell, Fig. 1(d), inserts two additional NMOS and PMOS transistors into a modified 6T latch in a configuration that breaks the direct feedback path responsible for SEU propagation [5]. Nodes A and C are rendered immune to SNU by the feedback structure, while nodes B and D retain finite Q_{crit} values. Write functionality requires that the pull-up ratio PR_2 — defined as the ratio of the PMOS width to the relevant NMOS pull-down width — be held at or below 0.75, which necessitates selectively upsizing specific NMOS devices rather than applying uniform minimum sizing. This intentional asymmetry has direct consequences for the SEU response of individual nodes.

III. SIMULATION METHODOLOGY

All simulations were performed in Cadence Virtuoso ADE L using the Spectre simulator with the GPDK 45 nm process design kit at $V_{DD} = 1.1$ V and $T = 27^\circ\text{C}$ unless otherwise noted. Schematics were constructed following the original published topologies: DICE from Calin *et al.* [4], Quatro from Jahinuzzaman *et al.* [5] (Fig. 3), and RSP from Peng *et al.* [6].

A. Transistor Sizing

All transistors were set to minimum dimensions of $W/L = 120/45$ nm for the initial architecture comparison, isolating the effect of cell topology on SEU hardening rather than transistor drive strength. Two topology-specific exceptions were required to faithfully reproduce the published designs.

1) *14T RSP Cell*: The four pull-down NMOS transistors (N0–N3) were sized to $W/L = 360/45$ nm, a 3:1 driver-to-pull-up width ratio specified in Peng *et al.* [6] to balance read stability and write ability while maximizing post-strike restoring current.

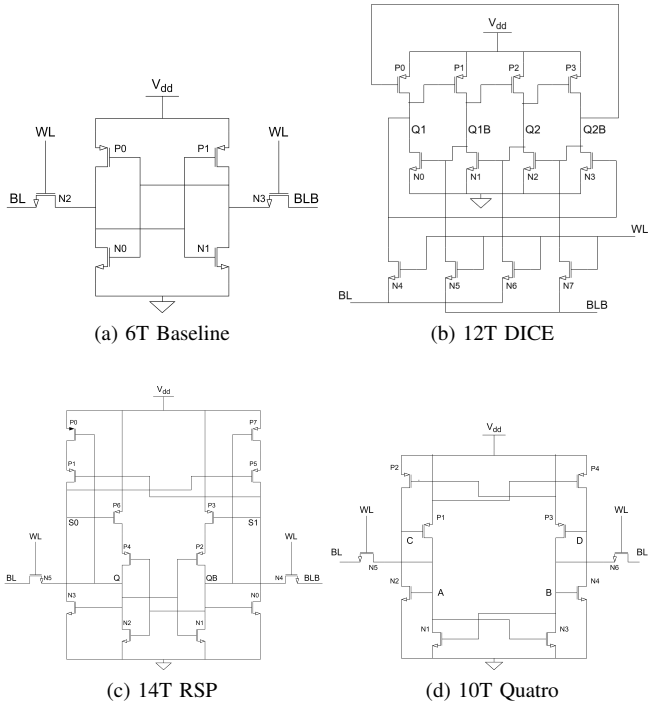


Fig. 1. Schematics of the four SRAM bitcell topologies evaluated in this work: (a) standard 6T baseline, (b) 12T DICE [4], (c) 14T RSP [6], and (d) 10T Quatro [5].

2) *10T Quatro Cell*: The two pull-down NMOS transistors on nodes C and D (N2 and N4) were sized to $W/L = 240/45$ nm. Uniform minimum sizing violates the pull-up ratio constraint $PR_2 \leq 0.75$ defined in Jahinuzzaman *et al.* [5], causing the cell to fail write operations entirely.

B. SEU Injection Protocol

SEU events were modeled using a double-exponential current source injected directly onto the target storage node, a well-established simulation methodology for reproducing the transient current profile of a particle strike [3]. The injected current follows the form:

$$I(t) = I_{\text{peak}} \left(e^{-(t-t_d)/\tau_2} - e^{-(t-t_d)/\tau_1} \right) \quad (1)$$

where $\tau_1 = 10$ ps and $\tau_2 = 150$ ps are the rise and fall time constants, $t_d = 1$ ns is the strike onset delay, and I_{peak} scales with the injected charge as $I_{\text{peak}} = Q_{\text{inj}}/140$ ps. All SEU simulations were conducted in hold mode with the wordline deasserted ($V_{WL} = 0$ V) and both bitlines precharged to V_{DD} , ensuring that the access transistors remained off and exerted no influence on the storage nodes during the strike event.

Initial conditions were enforced via Spectre convergence aids (nodesets). For cells storing a logic 1 at the primary storage node, the high nodes were initialized to V_{DD} and the complementary nodes to ground; polarity was reversed for the storing-0 case. Transient simulations were run for 5 ns with a maximum timestep of 1 ps and tightened solver tolerances ($\text{reltol} = 10^{-4}$, $\text{vabstol} = 10^{-7}$, $\text{iabstol} = 10^{-13}$) to ensure accurate resolution of the fast strike transient and reliable convergence near the Q_{crit} threshold.

C. Q_{crit} Extraction

Q_{crit} was extracted using an automated bisection algorithm implemented in the Cadence OCEAN scripting environment. Beginning with a search interval of $[2, 50]$ fC, each iteration evaluated the struck node voltage at $t = 5$ ns and updated the interval bounds depending on whether the cell had recovered or latched into the upset state. After eight bisection iterations the interval converged to a precision of approximately 0.19 fC, at which point the midpoint was recorded as Q_{crit} . A permanent upset was declared when the struck node voltage remained on the incorrect side of $V_{DD}/2$ at the end of the simulation window.

For cells with theoretical SNU immunity, single-node injection was verified to produce no upset across the full $[2, 50]$ fC search range, confirming the topological immunity property in simulation. DNU characterization was performed by injecting simultaneously onto both nodes of each candidate pair with equal charge magnitude.

D. Metrics

1) *Critical Charge*: Q_{crit} is reported separately for each sensitive node and polarity combination. For SNU characterization, charge is injected onto a single storage node while all remaining nodes are unperturbed. For DNU characterization, equal charge is injected simultaneously onto both nodes of each candidate pair.

2) *Static Noise Margin*: Hold-mode SNM was extracted from the DC butterfly curve. A voltage source V_n was swept from 0 to V_{DD} at the input of one inverter while the output voltage at the complementary node was recorded at 221 equally spaced points. The DC sweep was run with the `hysteresis` option enabled to correctly trace both branches of the bistable characteristic. SNM was computed in post-processing as the side length of the largest square inscribable within the eye opening of the resulting voltage transfer characteristic, following the standard butterfly curve methodology.

3) *Leakage Power*: Standby leakage power was measured via a DC operating point simulation with all injection sources zeroed and the wordline deasserted. Total leakage power was computed as $P_{\text{leak}} = |I_{V_{DD}}| \cdot V_{DD}$, where $I_{V_{DD}}$ is the quiescent current drawn from the supply source. Nodesets were applied to enforce the storing-1 state consistently across all PVT conditions.

4) *Read Access Time*: Read access time was measured at nominal conditions ($V_{DD} = 1.1$ V, $T = 27^\circ\text{C}$, TT corner) for the minimum-sized implementation of each topology. Both bitlines were precharged to V_{DD} with a 10 fF load capacitance, and access time was defined as the interval from wordline assertion to a 50 mV differential between BL and $\overline{\text{BL}}$. This metric was not included in the DICE parametric sweep, as it improves monotonically with transistor width and presents no meaningful optimization trade-off.

E. Parametric Sweep and Corner Analysis

Following the minimum-sized four-cell comparison, the 12T DICE cell was selected for comprehensive characterization across the full PVT design space. A master OCEAN script coordinated three independent sweep loops — SEU, leakage power, and SNM — executed sequentially across all combinations of the parameters listed in Table I.

The access transistor width was held fixed at the minimum $W/L = 120/45$ nm throughout, as it is electrically isolated from the storage nodes during hold-mode simulation and does not influence any of the three metrics under study. The SEU sweep iterated over both storage polarities, yielding $6 \times 6 \times 5 \times 5 \times 5 \times 2 = 9,000$ total bisection runs. The power and SNM sweeps omitted the polarity dimension, producing $6 \times 6 \times 5 \times 5 \times 5 = 4,500$ runs each, for a combined total of 18,000 simulations. Results from all three sweeps were imported into Python for post-processing, visualization, and sensitivity analysis.

TABLE I
PARAMETRIC SWEEP SPACE FOR DICE PVT CHARACTERIZATION

Parameter	Values	Points
$W_{n,\text{store}}$	120, 160, 200, 240, 280, 320 nm	6
$W_{p,\text{store}}$	120, 160, 200, 240, 280, 320 nm	6
Temperature	-40, 0, 27, 85, 125 °C	5
V_{DD}	1.00, 1.05, 1.10, 1.15, 1.20 V	5
Process corner	TT, FF, SS, FS, SF	5
Polarity (SEU only)	Storing 1, Storing 0	2

IV. RESULTS

A. 6T Baseline

The standard 6T cell serves as the unhardened reference for all comparisons. Under single-node injection, Q_{crit} was measured as 7.38 fC when striking the node storing a logic 1, and 10.14 fC when striking the node storing a logic 0. The asymmetry reflects the difference in restoring drive strength between the PMOS pull-up and NMOS pull-down at minimum sizing. Leakage power at nominal conditions was 25.4 pW, and read access time was 42.4 ps, establishing the area- and power-efficient baseline against which the hardened topologies are evaluated.

B. 12T DICE Cell

The DICE cell exhibited complete SNU immunity across the full injection range of 2 fC to 50 fC for all storage nodes and both polarities, confirming the topological nature of its hardening. This immunity is a structural property of the cell and does not depend on transistor sizing or physical layout.

DNU characterization was performed by injecting simultaneously onto the Q1+Q2 and Q1B+Q2B node pairs, which represent the critical DNU case. Q_{crit} was 7.32 fC for the pair storing logic 1 and 10.02 fC for the pair storing logic 0. These values closely match the 6T SNU Q_{crit} measurements, which is theoretically expected: with the access transistors off during hold mode, each DICE storage node is driven by the same pull-down or pull-up transistor dimensions as the corresponding node in the 6T cell, so per-node restoring strength — and therefore DNU Q_{crit} — is equivalent. DNU hardening in the DICE cell is layout-independent, as the critical pairs are separated by topology rather than physical placement.

Read access time improved significantly relative to the 6T baseline, measuring 25.1 ps, attributable to the additional drive provided by the interlocked feedback transistors during bitline discharge. Leakage power was 53.2 pW, a $2.1\times$ increase over the 6T cell consistent with the doubled transistor count.

C. 14T RSP Cell

The RSP cell achieved an SNU Q_{crit} of 28.33 fC, representing a $3.84\times$ improvement over the 6T baseline. This result was obtained with the four pull-down NMOS transistors (N0–N3) sized at $W/L = 360/45$ nm per the 3:1 driver-to-pull-up ratio specified in Peng *et al.* [6]. Due to the full symmetry of the RSP topology, a single storage condition characterizes both polarities.

DNU characterization revealed a strongly non-uniform hardening profile spanning three tiers. The best-case DNU pair (QB+S1) yielded $Q_{\text{crit}} = 28.45$ fC, essentially matching the SNU result, because S1 is a feedback-reinforced intermediate node that provides additional restoring current. Striking QB and the PM2 source node produced $Q_{\text{crit}} = 22.7$ fC, as the supply node is not a storage element and is difficult to perturb. The Q+S1 pair was significantly weaker at 5.24 fC. The worst-case pair, Q+QB, yielded $Q_{\text{crit}} = 3.77$ fC, which falls *below* the unhardened 6T SNU baseline of 7.38 fC. This critical vulnerability arises because the two main storage nodes share a common pull-down feedback path. When both are struck simultaneously, no independent restoring path remains active. Mitigating this vulnerability requires substantial physical separation of Q and QB in layout, making DNU hardening entirely layout-dependent for the RSP cell.

Leakage power was 94.6 pW, the highest of the four topologies, reflecting both the increased transistor count and the widened pull-down devices. Read access time was 34.4 ps.

D. 10T Quatro Cell

SNU characterization of the Quatro cell revealed a node-dependent hardening profile. Nodes A and C were confirmed immune to single-node upset across the full 2 fC to 50 fC injection range, consistent with the feedback structure described in Jahinuzzaman *et al.* [5]. Node B exhibited a finite Q_{crit} of 10.44 fC, and node D of 13.01 fC. The elevated Q_{crit} at node D relative to node B is a direct consequence of the intentional asymmetric sizing applied to satisfy the write-enable pull-up ratio constraint ($PR_2 \leq 0.75$).

DNU results are as follows: the A+B pair yielded $Q_{\text{crit}} = 5.03$ fC and was found to be polarity-independent, consistent with the symmetric roles of those nodes in the feedback topology. The B+C pair produced 7.32 fC, and the A+D pair 10.37 fC, with the higher value reflecting the additional drive strength at node D from the asymmetric sizing.

Leakage power was 60.2 pW and read access time was 43.9 ps, marginally slower than the 6T baseline due to the additional series transistors in the read discharge path.

E. Comparative Summary

Table II summarizes the worst-case Q_{crit} for each topology alongside transistor count, leakage power, and read access time. The DICE cell won in each evaluated metric and was consequently selected for further analysis.

TABLE II
PERFORMANCE SUMMARY — MINIMUM-SIZED CELLS, NOMINAL CONDITIONS ($V_{\text{DD}} = 1.1$ V, $T = 27^\circ\text{C}$, TT CORNER)

Architecture	Transistors	SNU Q_{crit} (fC)	DNU Q_{crit} , worst (fC)	DNU Q_{crit} , best (fC)	SNM (mV)	Leakage (pW)	Read t_{acc} (ps)
6T Baseline	6	7.38	—	—	370	25.4	42.4
12T DICE	12	∞	7.32	10.02	384	53.2	25.1
14T RSP	14	28.33	3.77	28.45	359	94.6	34.4
10T Quatro	10	10.44 (B) / 13.01 (D)	5.03 (A+B)	10.37 (A+D)	180.5	60.2	43.9

F. DICE Parametric Sweep

To evaluate the behavior of the DICE cell under realistic operating conditions, a comprehensive parametric sweep was performed across transistor sizing, temperature, supply voltage, and process corners. The resulting dataset was analyzed to identify key trends in radiation robustness and power consumption.

1) *Impact of Transistor Sizing on Q_{crit}* : Fig. 2 shows the sensitivity of Q_{crit} to storage transistor sizing. Increasing both NMOS and PMOS widths leads to a consistent increase in Q_{crit} , reflecting stronger restoring current at the storage nodes. The trend is monotonic across the explored design space, confirming that transistor sizing is a primary factor in determining SEU robustness.

2) *Temperature Dependence*: The effect of temperature on Q_{crit} is shown in Fig. 3. As temperature increases from -40°C to 125°C , Q_{crit} decreases significantly. This behavior is attributed to reduced carrier mobility and increased leakage at higher temperatures, which weaken the restoring capability of the cross-coupled inverters.

3) *Power–Robustness Trade-off and Optimal Design Point*: A trade-off between leakage power and SEU robustness is observed across the full PVT design space. Fig. 4a presents the Pareto frontier computed over all combinations of transistor sizing, temperature, supply voltage, and process corner, where each point represents a non-dominated configuration with respect to leakage power and Q_{crit} .

The identified knee point — the configuration at the boundary where further gains in Q_{crit} require disproportionate increases in leakage power — corresponds to $W_{n,store} = 320\text{ nm}$, $W_{p,store} = 120\text{ nm}$, at $T = -40^{\circ}\text{C}$ ($Q_{crit} = 29.7\text{ fC}$, $P_{leak} = 41.9\text{ pW}$). The dominance of cold-temperature configurations on the frontier is consistent with the temperature dependence shown in Fig. 3: lower temperatures suppress leakage and strengthen restoring currents simultaneously, placing cold-corner designs at the efficient boundary. In a space deployment

context, this result motivates active thermal management as a reliability strategy. Operating near the cold end of the thermal envelope yields the most favorable power–robustness trade-off for a given transistor sizing and should be prioritized.

4) *Multi-Objective Design Space Visualization*: This three-dimensional view further confirms that SEU robustness and SNM are both improved by stronger device sizing, but at the cost of increased leakage power, forming a coupled multi-objective design trade-off.

5) *SNM and Leakage Power Trade-offs*: Fig. 4c illustrates the butterfly curve derived from the inverter voltage transfer characteristics. The separation between the two curves reflects the stability of the storage nodes, and the minimum distance provides a measure of SNM.

This metric is used in the subsequent analysis to evaluate the trade-off between stability and leakage power across different design configurations.

V. DISCUSSION

Among the four architectures evaluated, the distinction between SNU and DNU performance proves to be the critical differentiator for space-grade memory selection. While all three hardened topologies offer meaningful SNU improvement over the 6T baseline, DNU Q_{crit} varies widely and exposes architecture-specific vulnerabilities that SNU metrics alone do not capture.

The RSP cell's $3.84\times$ SNU hardening is effectively negated under simultaneous Q+QB injection, where its worst-case DNU Q_{crit} of 3.77 fC falls below the unhardened 6T reference of 7.38 fC . This result underscores that SNU immunity is a necessary but insufficient criterion for space-grade memory selection, and that the RSP cell's high hardening ceiling is only realizable when layout enforces sufficient physical separation between the primary storage nodes.

The DICE cell's DNU Q_{crit} of 7.32 fC to 10.02 fC closely matches the 6T SNU values, which is theoretically consistent.

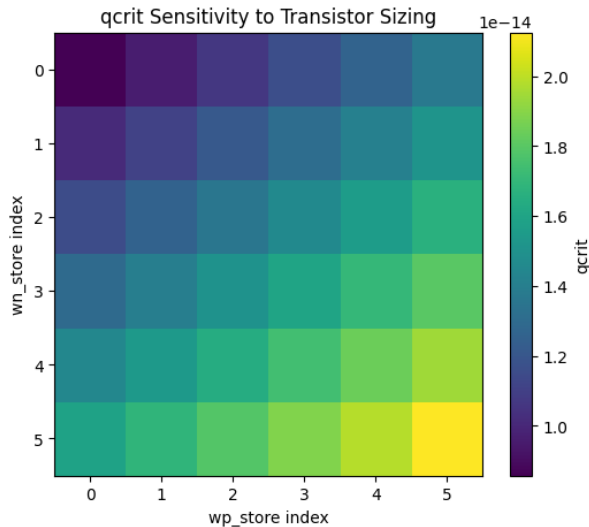


Fig. 2. Sensitivity of Q_{crit} to storage transistor sizing. Larger device widths improve restoring strength and increase critical charge.

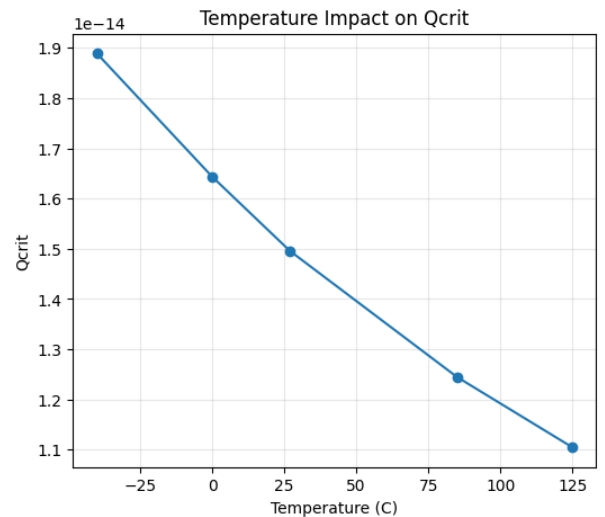


Fig. 3. Temperature dependence of Q_{crit} . Higher temperatures reduce restoring strength and increase vulnerability to upset.

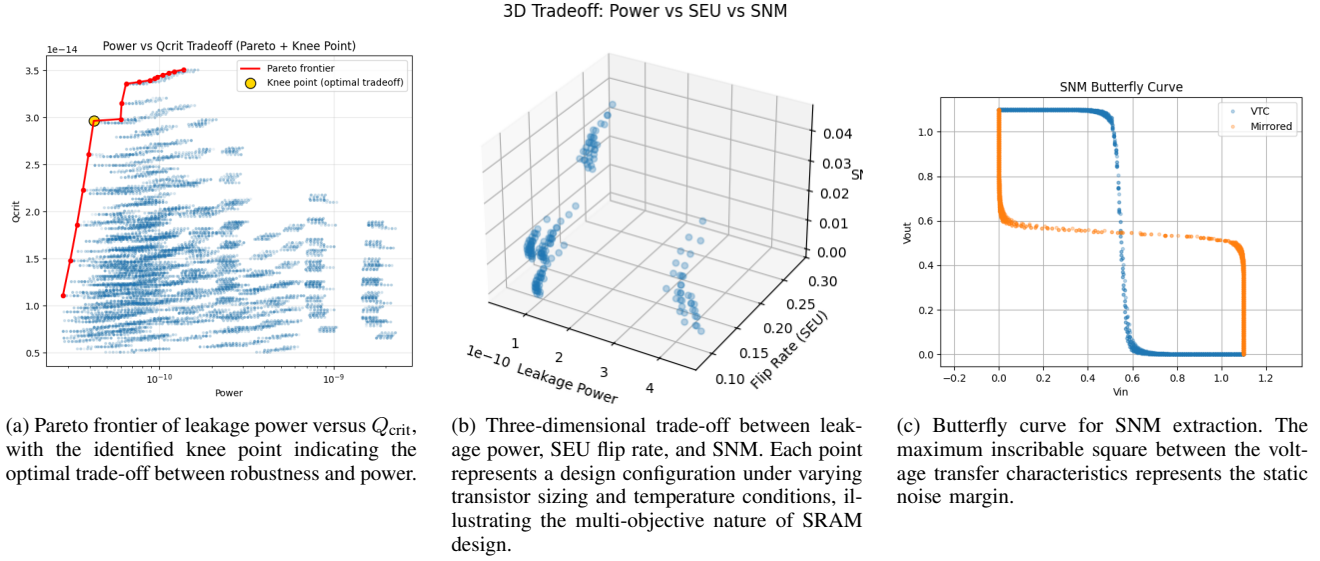


Fig. 4. DICE parametric sweep analysis. (a) Power–robustness Pareto frontier. (b) Three-dimensional multi-objective trade-off space across sizing and temperature. (c) Hold-mode SNM butterfly curve at nominal conditions.

Critically, DICE DNU hardening is layout-independent — the critical pairs are separated by topology rather than physical placement — making the cell’s hardening behavior predictable and transferable across layout implementations.

The Quatro cell occupies an intermediate position. Its partial node immunity (nodes A and C) and compact 10T footprint offer an attractive area–hardening trade-off for applications where physical layout can be rigorously optimized. However, the finite Q_{crit} at nodes B and D limits its suitability for sensitive applications.

The parametric sweep results confirm that transistor sizing and temperature are the dominant factors governing DICE radiation robustness. Increasing device width improves Q_{crit} monotonically but at the cost of leakage power, establishing a fundamental design trade-off. The Pareto frontier analysis provides a systematic framework for navigating this trade-off: the identified knee point at $W_{n,store} = 320$ nm, $W_{p,store} = 120$ nm represents the most efficient power–robustness balance within the explored design space.

The strong temperature dependence of Q_{crit} has direct implications for space deployment. The Pareto frontier is dominated by cold-corner configurations, reflecting the simultaneous suppression of leakage and strengthening of restoring currents at low temperature. However, in an industry application, the operative design constraint is the hot corner: transistor sizing must be selected to meet the Q_{crit} target at $T = 125^\circ\text{C}$, the worst-case thermal extreme. This framing motivates active thermal management as a system-level reliability strategy, as operating near the cold end of the LEO thermal envelope yields substantial gains in both power efficiency and radiation hardness that passive sizing alone cannot achieve.

Several limitations of this work should be noted. All DNU simulations assumed simultaneous equal-magnitude charge injection on both nodes, representing a worst case; in practice, charge sharing depends on particle track geometry and inter-

node spacing, neither of which is captured at the schematic level. Read access time measurements used a fixed 10 fF bitline capacitance without a sense amplifier model, limiting direct comparison with array-level implementations. Process variation was explored through corner analysis (TT, FF, SS, FS, SF) but not through statistical Monte Carlo sampling, which would provide Q_{crit} distributions rather than point estimates.

VI. CONCLUSION

This work evaluated four SRAM bitcell topologies for radiation hardness in the context of space-based AI inference, using SPICE-level simulation in the GPDK 45 nm process. Among the three hardened architectures, the 12T DICE cell emerged as the most suitable candidate, offering complete SNU immunity by topology and layout-independent DNU Q_{crit} that closely tracks the 6T SNU baseline. The RSP cell, despite achieving the highest SNU Q_{crit} at 28.33 fC, exhibits a worst-case DNU vulnerability of 3.77 fC — below the unhardened 6T reference — making its hardening entirely contingent on physical layout separation. The Quatro cell offers a compact intermediate option but is similarly limited by finite Q_{crit} at nodes B and D.

Parametric characterization of the DICE cell confirmed that transistor sizing and temperature are the dominant factors governing radiation robustness. Q_{crit} increases monotonically with device width at the cost of leakage power, while rising temperature degrades restoring strength significantly across the -40 to 125°C range. These findings highlight that active thermal management can yield compounding benefits in both power efficiency and radiation hardness. Together, these results provide a quantitative basis for SRAM architecture selection in radiation-sensitive space computing systems.

ACKNOWLEDGMENT

John Gannon implemented and simulated the 12T DICE and 10T Quatro bitcells, developed the SEU injection methodology, and designed and scripted the parametric simulations for sizing, supply voltage, temperature, and process corners. John was also the primary contributor to Sections I-IV.

Arnur Maratov implemented and simulated the baseline 6T and 14T RSP bitcells, performed cross-validate simulations for consistent results, SNU and DNU critical charge extraction across all node pairs, and measurement of leakage power and read/write access time. Arnur also contributed to Sections IV-VI.

Yoran Wong developed the Python-based post-processing workflow to analyze the simulation datasets, including data cleaning, metric extraction, and visualization. Yoran generated the parametric sweep figures, constructed the power- Q_{crit} trade-off analysis, and identified the Pareto frontier and corresponding knee point for optimal design selection.

All authors contributed to the overall project design, analysis, and editing of the final report.

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